Department of Information Technology

Academic Year 2023 – 2024(Odd Semester)

Degree, Semester & Branch: III Semester – B.Tech. Information Technology

Course Code & Title: CD3291 – Data Structures and Algorithms

Name of the Faculty member (s): Mrs. G. Mareeswari

Innovative Practice Description

- **Unit / Topic:** Unit III / Sorting
- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO9
- **Activity Chosen:** Flash Cards
- **Date of Implementation:** 30.10.2023
- **Justification:**
    - Flash card is a card bearing information which is intended to be used as an aid in memorization
    - Sorting is an important algorithm in Computer Science. These algorithms have direct applications in searching algorithms, database algorithms, divide and conquer methods, data structure algorithms, and many more. Sorting can often used to reduce the complexity of a problem
    - To have a good remembrance and understanding, the flash card activity is chosen for the topic "Sorting"
- **Time Allotted for the Activity**: 15 Minutes
- **Pre Implementation:**
    - Two types of Flash cards were prepared. One containing question and other card containing answers as shown in Fig 1.
    - Flash card with questions is shuffled with the cards containing answers
- **Details of the Implementation:**
    - Questions and answers were prepared in a card.
    - Question cards were distributed to certain group of students and answer cards were distributed to few group of students.
    - Student who is having the answer read out the card. Other students listed to the question and found out the question for that answer.
    - Student who is having the respective question card explained the question and answer for the question to the entire class as shown in Figure 2

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- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO3	**2**	**1**	**1**	**1**	**1**	**1**

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
Justification for correlation	Students will be able to understand the concept of Sorting	Students will be able to choose the appropriate Sorting algorithm to approach the problem	Students will be able to provide solution for the problem by applying different sorting algorithms.	As the student involved in the activity as a team of two, the team work skill improved	Students communication skill will be improved as they are discussing the answers with peers	The problem solving skill earned through this activity helps the students in solving various problems

- **Images / Screenshot of the practice:**

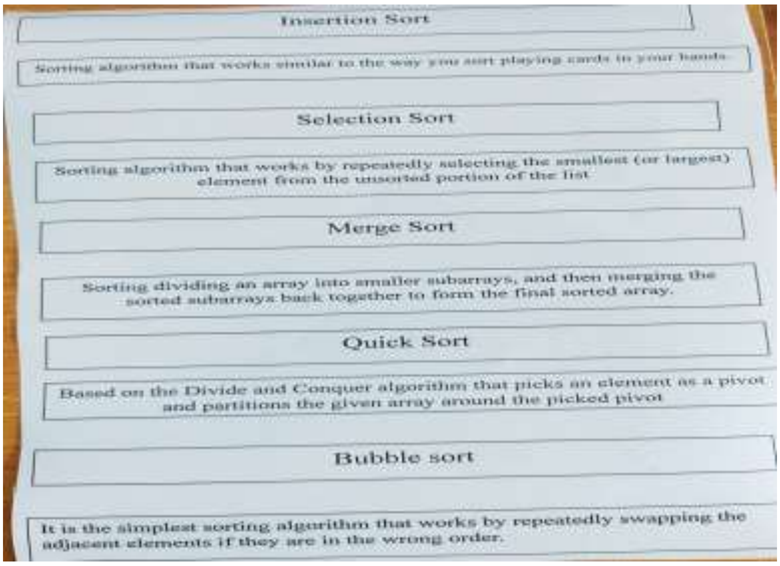

Fig 1: Sample of Flash cards

Fig 2: Sample of Students (Gowtham. C) involvement in Flash Card Activity

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - Students said that the flash card activity is very joyful to recollect the Sorting - Bubble, Insertion, Selection, Quick and Merge sort concepts.
    - They asked to conduct more of the similar activity during class which help them to listen the class without getting bored.
  - ***Benefit of the practice:***
    - Students understanding are improved for the concepts of different types of sorting algorithms.
    - Students memorization and remembrance is improved
  - ***Challenges faced in implementation:***
    - Few students found it difficult to find the corresponding flash card which containing answer.

References:

- https://en.wikipedia.org/wiki/Flashcard
- https://ellii.com/blog/10-ways-to-use-flashcards-in-class
- https://www.ritrjpm.ac.in/images/computerscience/13_CS6703_Falshcard
- https://www.teachingenglish.org.uk/article/using-flash-cards-young-learners

Signature of Faculty Member **HOD-IT**